

How TWS works

The radar uses a frequency-modulated continuous wave (FMCW) radar architecture operating at 1-4GHz band, with good range resolution of 5cm due to its high-bandwidth chirp. A narrow-band, spatial-frequency filter provides a range gate that eliminates the wall from the image, facilitating maximum receiver dynamic range to be applied to the target scene behind the wall. A time division multiplexed (TDM), multiple-input multiple-output (MIMO) array provides a low-cost, least-complicated solution to a fully-populated antenna aperture capable of near-field imaging.

To achieve 16-frames-per-second video-frame-rate imaging, a data-pipeline and streamlined imaging algorithm is developed. Coherent frame-to-frame processing rejects stationary clutter, revealing the location of moving targets. Maximum expected range through a solid concrete wall is 20 metres.

Advantages of through-wall radar technology

1. The system enables quick tactical decision making in urban operations.
2. It allows rapid determination of the presence and location of multiple people behind walls.
3. Both static and moving objects can be displayed simultaneously. This helps when the user needs to understand the layout of the place to plan for informed and effective operations.
4. The system can sense through many type of walls including cement, dry wall, brick, plaster and reinforced concrete.
5. The system supports remote display over secure Wi-Fi protocol, which makes it convenient for several users to access the information simultaneously for tactical operation.
6. The system is portable, lightweight and can withstand up to three hours of continuous preparation.

Current limitations of TWS technology

1. Buildings that have solid metal walls or insulation with foil backing are a problem for radar based TWS devices.
 2. Most technologies provide a lower-resolution image as compared to normal video images. Sometimes, it gets difficult to differentiate between items that are displayed on the screen, for example, distinguishing between a mobile phone and a handheld pistol.
 3. A full-fledged system that offers actual video identifying moving people, static objects and room structure is very expensive.
- TWS technology is continuously evolving and improving. With improving capabilities, lowered costs and increasing availability of TWS technology, it will be easier to save lives by reducing the risk to law-enforcement officers in tactical situations so that officers can make quicker, smarter, life-saving decisions. ●



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